PHYSICS

14·1·2024

1. (2) 3 : 4
2. (1) Rs. 360
3. (1) convergent lens

$l = 3x \quad x$

$d = 2y \quad y$

$r = \frac{d^2}{4}$

$A = \pi r^2 = \frac{\pi d^2}{4}$

$A_1 = \frac{\pi 4y^2}{4}$

$A_2 = \frac{\pi y^2}{4}$

$\frac{R_1}{R_2} = \frac{\frac{\rho 3x}{\frac{\pi 4y^2}{4}}}{\frac{\rho x}{\frac{\pi y^2}{4}}} = \frac{\frac{4\rho 3x}{\pi 4y^2}}{\frac{4\rho x}{\pi y^2}} = \frac{\frac{\rho 3x}{\pi y^2}}{\frac{4\rho x}{\pi y^2}} = \frac{\rho 3x}{\pi y^2} \times \frac{\pi y^2}{4\rho x} = \frac{3}{4}$

$100W = 0.1 kW$

$4 \times 0.1 = 0.4 kW \times 6h = 2.4 kWh \times 30$

$= 72.0 kWh \times 5$

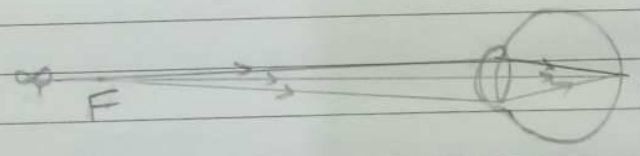
$= ₹360$

4. The two common defects of vision are :-

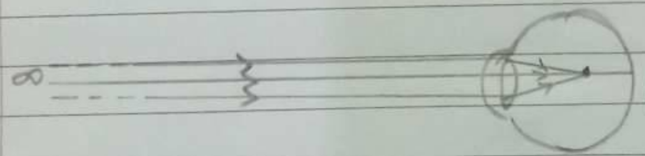
i) Myopia (near-sightedness) :-

★ In this defect a person can see the nearby objects clearly but cannot see far-off objects clearly

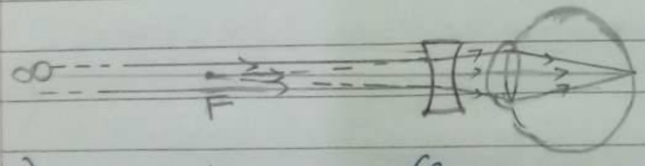
★ Correction – Using a concave lens of suitable ~~focal length~~ power



Far point of myopic eye
(Far point is less than infinity)



Myopic eye
(Image is formed in front of retina)

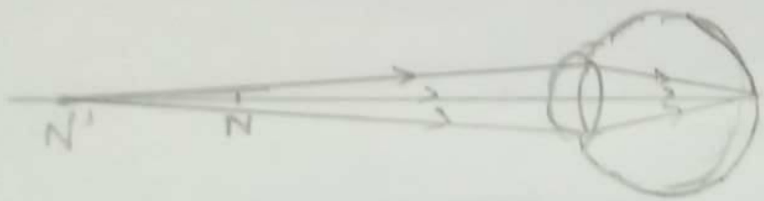


Correction of the myopic eye using concave lens

ii) Hypermetropia (far-sightedness) :-

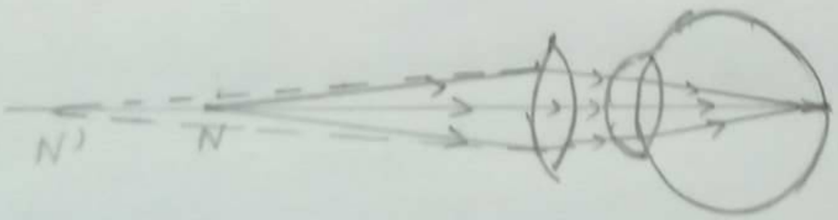
★ In this defect a person can see the far-off objects clearly but cannot see nearby objects clearly

★ Correction – Using a convex lens of a suitable power



Near point of a hypermetropic eye
(Near point is more than 25cm)

Hypermetropic eye
(Image is formed behind the retina)



Correction of a hypermetropic eye using convex lens

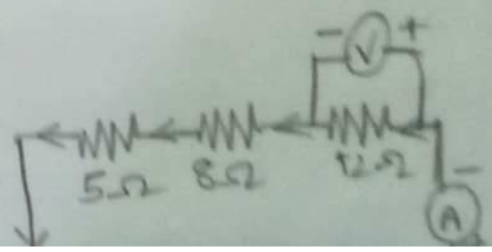
5.

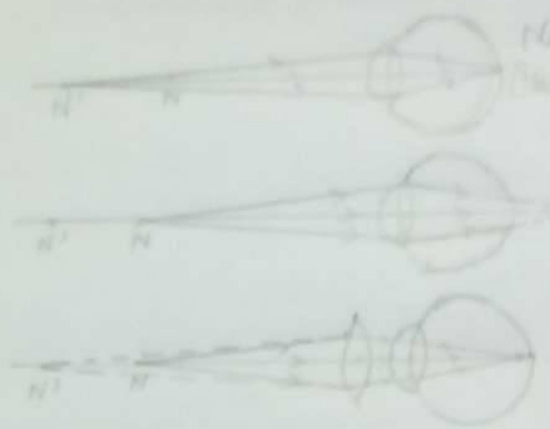
i) $R_s = R_1 + R_2 + R_3 = 5 + 8 + 12$

$= 25\,\Omega$

$V = 2 + 2 + 2 = 6V$

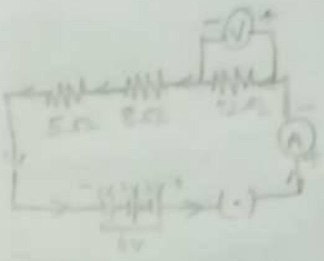
6/25





Correction of a Hypermetropic eye using convex lens

5.
i) $R_S = R_1 + R_2 + R_3 = 5 + 8 + 12$
$= 25\,\Omega$



$V = 2+2+2 = 6V$

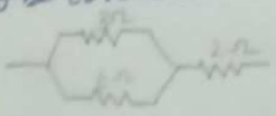
By ohm's law, $V = IR \Rightarrow I = \frac{V}{R} = \frac{\cancel{25}\ 6/25}{\cancel{6}} = \cancel{\ldots}$ $0.24A$

Potential difference across $12\,\Omega$ resistor,

$V = IR = \cancel{\frac{25(12)}{6}}\ (0.24)(12)$
$= 2.88V$

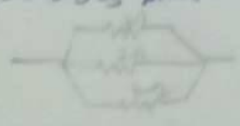
∴ The reading in the ammeter is $0.24A$ & the reading in the voltmeter is $2.88V$

ii)(b)(1) To give a total resistance of $4\,\Omega$, the 3Ω & 6Ω resistors have to be connected in parallel and this combination has to be connected in series as,



$R = \frac{\cancel{3}\times\cancel{6}^{2}}{\cancel{9}_{\cancel{3}}} + 2 = 2+2 = 4\,\Omega$

(2) To give a total resistance of 1Ω, the 3Ω, 6Ω & 2Ω resistors have to be connected in parallel as,



$R = \frac{1}{3} + \frac{1}{2} + \frac{1}{6} = \frac{2+3+1}{6} = \frac{6}{6} = 1\,\Omega$

6.
(i)(2)(a) Given, $u = -25cm$, $v = -100cm$

By lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ $\quad \frac{1}{-100} - \frac{1}{-25} = \frac{1}{f} \Rightarrow \frac{-1 + 4}{100} = \frac{1}{f}$

$\Rightarrow \frac{3}{100} = \frac{1}{f} \Rightarrow f = \frac{100}{3}cm$
$= 33.33cm$

∴ The focal length of the lens is 33.33 cm

$$P = \frac{1}{f} = \frac{1}{\frac{\frac{100}{3}}{100}} = \frac{1}{\frac{100}{3} \times \frac{1}{100}} = \frac{1}{\frac{1}{3}} = \frac{1}{1} \times \frac{3}{1} = 3D$$

∴ The power of the lens is +3D.

The lens is a convex lens.

(ii) * The ability of the eye lens to change its focal length so as to view both near and far off objects clearly is called accomodation of eye.

* When we see a near by object, our cilliary muscles contracts and focal length of the lens decreases.

* When we see a far off object, our cilliary muscles reelaxes and focal length of the lens increases

* A normal eye can see objects placed between 25cm and infinity

(iii) The iris controls the size of pupil. Pupil controls the amount of light entering our eye. Iris is a dark muscular diaphragm that ~~contr~~ imparts the colour of the eye

-1-2014

CHEMISTRY

1. (3) (i), (ii), (iii) only
2. (2) Sodium Zincate and hydrogen gas
3. (3) Aluminium

4. Since compound XY is formed by gaining and losing of electrons, the type of bonding between elements in compound XY is [illegible]